

Class 17 pt.2

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Q13. Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
dat <- read.table("rs8067378_ENSG00000172057.6.txt", header = TRUE)

# Inspect column names
names(dat)
```

```
[1] "sample" "geno"   "exp"
```

```
table(dat$genotype)
```

```
< table of extent 0 >
```

```
tapply(dat$exp, dat$geno, median)
```

```
      A/A      A/G      G/G
31.24847 25.06486 20.07363
```

```
# 1. Loading the data
dat <- read.table("rs8067378_ENSG00000172057.6.txt", header = TRUE)

# Checking column names (should be: "sample" "geno" "exp")
names(dat)
```

```
[1] "sample" "geno"   "exp"
```

```
# Making sure genotype is an ordered factor
dat$geno <- factor(dat$geno, levels = c("A/A", "A/G", "G/G"))

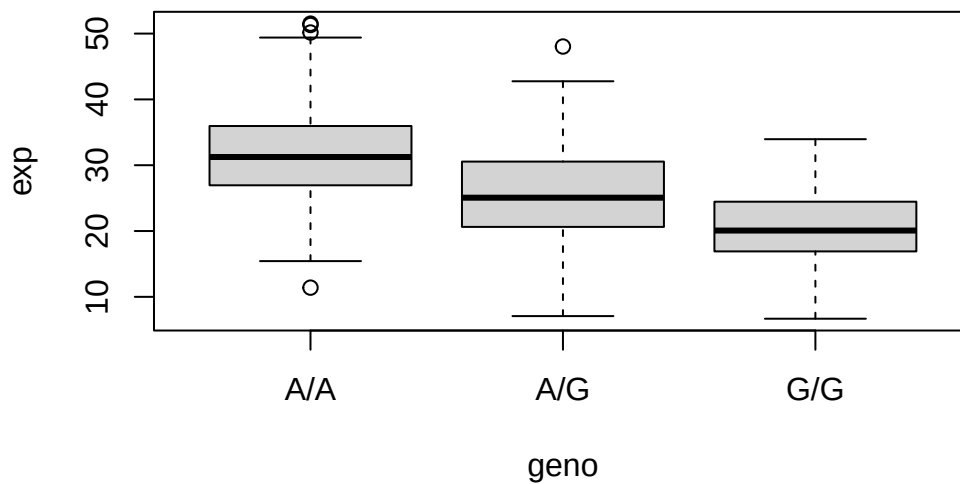
# 2. Sample size for each genotype
table(dat$geno)
```

```
A/A A/G G/G
108 233 121
```

```
# 3. Median expression for each genotype
tapply(dat$exp, dat$geno, median)
```

```
      A/A      A/G      G/G
31.24847 25.06486 20.07363
```

```
# 4. I preferred boxplot object to see five-number summary
bp <- boxplot(exp ~ geno, data = dat)
```



```
bp$stats # row 3 of this matrix = medians for each genotype
```

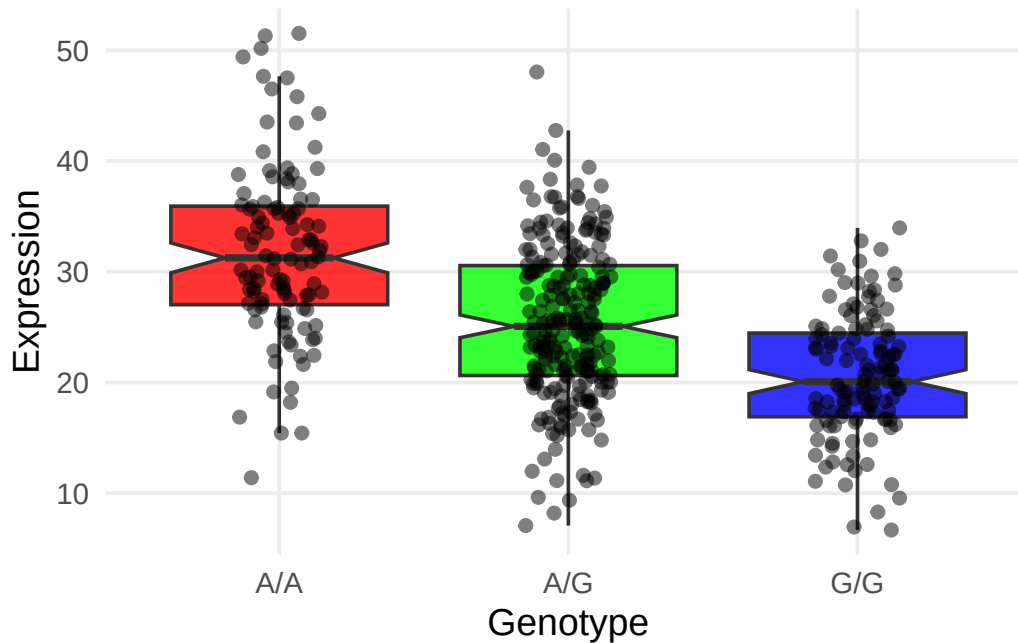
```
      [,1]      [,2]      [,3]
[1,] 15.42908  7.07505  6.67482
[2,] 26.95022 20.62572 16.90256
[3,] 31.24847 25.06486 20.07363
[4,] 35.95503 30.55183 24.45672
[5,] 49.39612 42.75662 33.95602
```

Q14 Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)

# Making sure geno is an ordered factor
dat$geno <- factor(dat$geno, levels = c("A/A", "A/G", "G/G"))

ggplot(dat, aes(x = geno, y = exp, fill = geno)) +
  geom_boxplot(
    notch = TRUE,
    outlier.shape = NA,
    alpha = 0.8
  ) +
  geom_jitter(
    width = 0.15,
    alpha = 0.5,
    size = 1.8
  ) +
  scale_fill_manual(values = c("red", "green", "blue")) +
  labs(
    x = "Genotype",
    y = "Expression"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    panel.grid.minor = element_blank()
  )
```



The boxplot shows clear differences in ORMDL3 expression across genotypes. Individuals with the A/A genotype have the highest expression values, A/G individuals have intermediate expression, and G/G individuals have the lowest expression. This pattern could indicate a dose-dependent effect of the SNP on ORMDL3 expression: each copy of the G allele reduces expression. Therefore, yes this SNP affects the expression of ORMDL3, with the G allele associated with decreased expression.